

Summary

Engineer and researcher with a PhD in Engineering Science (Mechanical Engineering). Experienced in **applied robotics** research at **KU Leuven** and **Flanders Make**. Specialized in the analysis and development of **data representations** (of motion and force trajectories), with a strong focus on **mathematical modeling**, **numerical robustness**, and dealing with edge cases and **imperfect data**. Trained in solving complex problems through **analytical reasoning** and **clear communication** within **multidisciplinary research teams**.

Professional Experience

2025–Present **Postdoctoral Researcher — Robotics Research Group and Flanders Make**
KU Leuven, Leuven, Belgium

- Continued research and formalization of invariant representations of rigid-body motion and interaction force trajectories, resulting in technical academic publications.
- Co-developed research project proposals; authored conference and workshop papers; presented research at international conferences.

2020–2025 **Doctoral Researcher — Robotics Research Group and Flanders Make**
KU Leuven, Leuven, Belgium

- Developed novel invariant representations for rigid body motion for robustly comparing motion trajectory data under changes in reference frame and measurement noise.
- Interpreted experimental and simulated data, with particular attention to edge cases, representation singularities, and model limitations.
- Researcher within the European research project *Robotgenskill*. Focused on robot skill learning and generalization by leveraging invariant representations and numerical optimal control.
- Researcher within the Flemish research project *Prorob*. Focused on spray paint deposition process modeling and simulation, robot trajectory generation, and process optimization for robotic spray painting in industrial settings.
- Worked within multidisciplinary teams, providing analytical and methodological input.
- (Co-)mentored students, guiding reasoning, methodology, and interpretation of results.
- (Co-)authored peer-reviewed journal and conference papers.
- Presented papers at international conferences.
- Assisted with undergraduate teaching and evaluation.

Degrees

- 2025 **PhD in Engineering Science (Mechanical Engineering)**, *KU Leuven*, Leuven, Belgium.
Thesis: *Invariant Trajectory Similarity Measurement: Resolving Singularity Issues for Robust Rigid-Body Motion Recognition*
- 2020 **Master of Science in Mechanical Engineering**, *KU Leuven*, Leuven, Belgium.
Specialization in Mechatronics and Robotics. Graduated *Cum Laude*.
(Master's thesis: *Summa Cum Laude*)
- 2018 **Bachelor of Science in Engineering (Specialisation: Electrical Engineering)**, *KU Leuven*, Leuven, Belgium.
Graduated *Cum Laude*.

Engineering and Analytical Competencies

- Mathematical modeling and interpretation
- Numerical Simulation development
- Numerical optimization and optimal control
- Linear algebra, differential geometry
- Technical documentation and clear explanation of methods and assumptions

Programming and Tools

- **MATLAB** and **Python**: numerical computing, modeling, simulation development, algorithm prototyping
- **ROS** and **C++**: algorithm implementation
- **Git CI/CD**: version control

Languages

Dutch	Native	<i>Mother tongue</i>
English	Fluent	<i>Used in academic and professional settings</i>
French	Conversational	<i>Can understand and communicate in routine situations.</i>

Award(s)

Best Poster Award at the 2023 Flanders Make Scientific conference on machines, vehicles, and production technology.

References

Available upon request.